

REAL SOURCE DOCUMENT — MIT COURSE CATALOG

Institution: Massachusetts Institute of Technology (MIT)

Program: Artificial Intelligence and Decision Making (Course 6-4) — Bachelor of Science

Source URL: <https://catalog.mit.edu/degree-charts/artificial-intelligence-decision-making-course-6-4/>

Supporting PDF: <https://catalog.mit.edu/degree-charts/artificial-intelligence-decision-making-course-6-4/artificial-intelligence-decision-making-course-6-4.pdf>

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Department: Electrical Engineering and Computer Science (EECS)

Note: Course 6-4 (AI and Decision Making) was introduced as a new degree program at MIT starting academic year 2022–2023.

COURSE 6-4: ARTIFICIAL INTELLIGENCE AND DECISION MAKING

PROGRAM OVERVIEW

The Bachelor of Science in Artificial Intelligence and Decision Making (Course 6-4) is offered by the Department of Electrical Engineering and Computer Science. This major covers the theoretical foundations and practical applications of artificial intelligence, machine learning, and decision-making under uncertainty.

GENERAL INSTITUTE REQUIREMENTS (GIRs)

Same GIR requirements as Course 6-3. See MIT_6003_CS_program_requirements.txt for GIR details.

DEPARTMENTAL PROGRAM — FOUNDATION SUBJECTS

Same introductory subjects as Course 6-3:

- 6.100A: Introduction to Computer Science Programming in Python
- 6.1200[J]: Mathematics for Computer Science (Prereq: Calculus I GIR)
- One introductory EECS lab subject (6.1010, 6.1020, or 6.1800)

DEPARTMENTAL PROGRAM — CORE AI SUBJECTS (all required)

- 6.1210: Introduction to Algorithms — Prereq: 6.100A + 6.1200[J]
- 6.1910: Computation Structures — Prereq: 6.1010 or equivalent
- 6.3700: Introduction to Probability — Prereq: Calculus II (18.02 GIR) [Note: alternative probability courses 6.3800, 18.05, 18.600 may substitute — check current catalog]
- 6.3900: Introduction to Machine Learning — Prereq: 6.1210 and (6.3700 or 18.600 or 18.05); covers supervised/unsupervised learning, neural networks, statistical learning theory
- 6.4100: Artificial Intelligence — Prereq: 6.1210; covers search, constraint satisfaction, Bayesian networks, reinforcement learning, natural language processing basics
- 6.4110: Representation, Inference, and Reasoning in AI — Prereq: 6.1210 and 6.3700; covers probabilistic graphical models, inference algorithms, decision theory

Note on prerequisite chains for 6-4 students:

To reach 6.3900 (Introduction to Machine Learning), the prerequisite chain is:

- 18.01 (Calculus I) and 18.02 (Calculus II) [GIRs, no MIT prereqs beyond high school math]
- Concurrent with 18.02: 6.100A (no prereq)
- After 18.01: 6.1200[J] (Mathematics for CS)
- After 6.100A + 6.1200[J]: 6.1210 (Introduction to Algorithms)
- After 18.02: 6.3700 (Introduction to Probability)
- After 6.1210 + 6.3700: 6.3900 (Introduction to Machine Learning)

Minimum path: 3 semesters from start

DEPARTMENTAL PROGRAM — DECISION SCIENCE SUBJECTS

- 6.4130: Bayesian Inference and Estimation — Prereq: 6.3700 and 6.3900
- 6.4140: Deep Learning — Prereq: 6.3900; covers neural network architectures, training, CNNs, RNNs, transformers

ADVANCED UNDERGRADUATE SUBJECTS (AUS)

6-4 students must select 2 AUS from the approved list, at least 1 from the Independent Inquiry list. Approved AUS for 6-4 include subjects in the 6.4xxx and 6.5xxx range related to AI, ML, robotics, and decision-making.

SUBJECTS WITH "PREREQUISITE OUTSIDE PROGRAM" NOTE

Some subjects appearing in the 6-4 degree chart are marked "Subject has prerequisites that are outside the program." This means the prerequisite subject is not itself required for 6-4 but must still be completed before enrollment. Students should consult the EECS advisor to plan around these requirements.

COMMUNICATION REQUIREMENT

6-4 students must complete 2 CI-M subjects from the departmental list.

SOURCE NOTES

Course 6-4 was established in AY2022–2023. The course description and requirements reflect the current MIT catalog (AY2025–2026). Students who enrolled before Fall 2022 in a related EECS program should consult the EECS Undergraduate Office for transition pathways.